Pricing Toolbox: A Guided Refresher

Itô calculus, Girsanov, path dependence, and early exercise

Lecture 14

Four tools solve four pricing questions

The question

How do the core tools of continuous-time finance work together when a payoff depends on a path, a barrier, or an exercise decision?

1. Use Itô's formula to handle nonlinear functions of Brownian motion.
2. Use Girsanov to move a drifted hitting problem to Brownian motion.
3. Augment the state for an Asian option and price it by PDE or simulation.
4. Use backward induction to decide whether an American option is exercised.

NO-ARBITRAGE Each technique changes the representation of a problem while preserving the economic object that must be priced.

Itô calculus turns stochastic integrals into identities

The quadratic-variation term is the essential correction

For $f(x) = e^x$, Itô's formula gives

$$\mathrm{d}e^{B_t} = e^{B_t}\,\mathrm{d}B_t + \frac{1}{2}e^{B_t}\,\mathrm{d}t.$$

Integrating from 0 to t,

$$\boxed{\int_0^t e^{B_s}\,\mathrm{d}B_s = e^{B_t} - 1 - \frac{1}{2}\int_0^t e^{B_s}\,\mathrm{d}s.}$$

Why ordinary calculus fails

The term $\frac{1}{2}f''(B_t)\,\mathrm{d}t$ comes from $(\mathrm{d}B_t)^2 = \mathrm{d}t$. Omitting it produces the wrong stochastic integral.

A quick Itô check

YOUR TURN Suppose someone claims

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What term is missing, and why?

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What term is missing, and why?

Answer

The correction is

$$-\frac{1}{2}\int_0^t e^{B_s}\,\mathrm{d}s.$$

It comes from the second derivative of e^x and Brownian quadratic variation.

The stochastic exponential is a martingale after correction

For deterministic square-integrable f, define

$$\mathcal{E}_t = \exp\left(\int_0^t f(s)\,\mathrm{d}B_s - \frac{1}{2}\int_0^t f(s)^2\,\mathrm{d}s\right).$$

Then

$$\mathrm{d}\mathcal{E}_t = \mathcal{E}_t f(t)\,\mathrm{d}B_t,$$

so $\mathcal{E}$ is a martingale under standard integrability conditions.

NO-ARBITRAGE The compensating term $\frac{1}{2}\int f^2$ is exactly what makes an exponential Gaussian quantity have mean one. It will become the density process in Girsanov.

Girsanov removes drift by changing measure

A change of measure turns drifted Brownian motion into Brownian motion

Let

$$X_t = \mu t + \sigma B_t, \qquad \theta = \frac{\mu}{\sigma},$$

under $\mathbb{P}$. Define

$$Z_t = \exp\left(-\theta B_t - \frac{1}{2}\theta^2 t\right), \qquad \left.\frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}}\right|_{\mathcal{F}_t} = Z_t.$$

Girsanov's theorem says

$$\widetilde{B}_t = B_t + \theta t$$

is a Brownian motion under $\mathbb{Q}$. Therefore

$$\boxed{X_t = \sigma \widetilde{B}_t \quad \text{under } \mathbb{Q}.}$$

The measure change simplifies a hitting-time question

Let

$$\tau_a = \inf\{t \geq 0 : X_t = a\}, \qquad a > 0.$$

Under $\mathbb{Q}$, τ_a is the first time a standard Brownian motion reaches a/σ. Its density is

$$f_{\tau_a}^{\mathbb{Q}}(t) = \frac{a/\sigma}{\sqrt{2\pi t^3}} \exp\left(-\frac{(a/\sigma)^2}{2t}\right).$$

Returning to the original measure

Use

$$\mathbb{P}(A) = \mathbb{E}^{\mathbb{Q}}[Z_t^{-1}\mathbf{1}_A],$$

then evaluate the density weight on the event $\{\tau_a = t\}$.

A drifted hitting time has an inverse-Gaussian density

The change-of-measure calculation yields

$$\boxed{f_{\tau_a}^{\mathbb{P}}(t) = \frac{a}{\sigma\sqrt{2\pi t^3}} \exp\left(-\frac{(a-\mu t)^2}{2\sigma^2 t}\right), \qquad t > 0.}$$

YOUR TURN When $\mu > 0$, does the process tend to hit a positive level sooner or later than a driftless Brownian motion?

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Answer

Sooner. Positive drift shifts the density toward earlier hitting times. The formula makes this visible through the term $(a - \mu t)^2$.

Martingale pricing with path-dependent payoffs

An Asian payoff needs a memory variable

An arithmetic-average Asian call pays

$$\left(\frac{1}{T} \int_0^T S_u \, \mathrm{d}u - K \right)^+ .$$

Define the running integral

$$I_t = \int_0^t S_u \, \mathrm{d}u.$$

At time t,

$$\int_0^T S_u \, \mathrm{d}u = I_t + \int_t^T S_u \, \mathrm{d}u.$$

NO-ARBITRAGE The stock price alone is not enough. The minimal Markov state is $\boxed{(t, S_t, I_t)}$.

State augmentation restores a pricing PDE

Under Black–Scholes,

$$\mathrm{d}S_t = rS_t\,\mathrm{d}t + \sigma S_t\,\mathrm{d}B_t, \qquad \mathrm{d}I_t = S_t\,\mathrm{d}t.$$

For $v(t, s, i)$, the pricing PDE is

$$\boxed{v_t + rs\,v_s + s\,v_i + \tfrac{1}{2}\sigma^2 s^2 v_{ss} - rv = 0,}$$

with terminal condition

$$v(T, s, i) = \left(\frac{i}{T} - K\right)^+.$$

What changed from a vanilla option

The new term $s\,v_i$ reflects the deterministic accumulation of the running integral; there is no diffusion term in the i-direction.

Monte Carlo prices the same Asian claim path by path

On a grid $t_k = k\Delta t$, simulate risk-neutral stock paths and form

$$\bar{S}_T^{(m)} = \frac{1}{T} \sum_{k=0}^{n-1} S_{t_k}^{(m)} \Delta t.$$

The price estimator is

$$\boxed{V_0 \approx e^{-rT} \frac{1}{M} \sum_{m=1}^{M} \left(\bar{S}_T^{(m)} - K \right)^+.}$$

YOUR TURN Where should the deterministic discount factor appear?

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Answer

Once outside the average. The simulated quantity is the risk-neutral payoff.

A geometric Asian option can reduce Monte Carlo noise

The geometric average

$$G = \exp\left(\frac{1}{T}\int_0^T \log S_u \,\mathrm{d}u\right)$$

is tractable in Black–Scholes because $\log S$ is Gaussian and its time integral is Gaussian.

Arithmetic Asian

The desired payoff uses the arithmetic average; it generally has no comparable closed form.

Control variate

Simulate both payoffs and use the known geometric-Asian value to reduce variance in the arithmetic-Asian estimator.

Barrier events combine a path maximum and Girsanov

A one-touch option is a hitting probability

Under $\mathbb{Q}$, let the stock pay dividend yield q:

$$\frac{\mathrm{d}S_t}{S_t} = (r - q)\,\mathrm{d}t + \sigma\,\mathrm{d}B_t.$$

A one-touch pays one unit at T if it ever reaches $H > S_0$:

$$\Pi = \mathbf{1}_{\{\sup_{0 \le t \le T} S_t \ge H\}}.$$

Log transform

With $X_t = \log(S_t/S_0) = \mu t + \sigma B_t$, where $\mu = (r - q) - \frac{1}{2}\sigma^2$, the barrier becomes the level $a = \log(H/S_0)$. Price the resulting hitting probability and discount it.

Reflection plus Girsanov gives the one-touch probability

For $Y_t = \mu t + \sigma B_t$,

$$\mathbb{Q}\left(\sup_{0\le t\le T} Y_t \ge a\right) = \Phi\left(\frac{\mu T - a}{\sigma\sqrt{T}}\right) + e^{2\mu a/\sigma^2}\Phi\left(\frac{-\mu T - a}{\sigma\sqrt{T}}\right).$$

When $\mu = 0$

Reflection gives twice the upper tail of B_T.

When $\mu \neq 0$

Girsanov removes drift, then the density change restores the original probability weights.

Worked one-touch: check the scale of a barrier probability

Let $S_0 = 100$, $H = 120$, $r = 5\%$, $q = 0$, $\sigma = 20\%$, $T = 1$. Then

$$\mu = 0.03, \qquad a = \log(1.2) = 0.1823,$$

and the hitting probability is approximately

$$\mathbb{Q}\left(\sup_{t \le 1} S_t \ge 120\right) \approx 0.413.$$

Therefore

$$\boxed{V_0 \approx e^{-0.05}(0.413) = 0.393.}$$

NO-ARBITRAGE A terminal digital and a one-touch are not the same contract: the barrier may be hit and then the stock may fall before maturity.

Discrete monitoring understates continuous barrier hitting

A simulation on dates $t_0, \ldots, t_N$ replaces

$$\mathbf{1}_{\{\sup_{0 \le t \le T} S_t \ge H\}}$$

by

$$\mathbf{1}_{\{\max_{0 \le k \le N} S_{t_k} \ge H\}}.$$

YOUR TURN For an up-and-in one-touch, is this discretely monitored estimator biased high or low relative to continuous monitoring?

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Answer

Low. A path can cross the barrier between two monitoring dates and return below it before the next observation. Refining the grid reduces this bias.

American options turn pricing into a decision

American pricing is backward induction with an exercise test

In a binomial model with gross return $R = 1 + r$ and risk-neutral probability p,

$$\boxed{V_t = \max\left\{ g(S_t),\ \frac{pV^u_{t+1} + (1-p)V^d_{t+1}}{R} \right\}.}$$

Exercise value

$$g(S_t).$$

Take the payoff and stop.

Continuation value

$$R^{-1}\mathbb{E}^{\mathbb{Q}}[V_{t+1} \mid \mathcal{F}_t].$$

Keep all future exercise choices.

A three-step American put has a genuine exercise region

Consider

$$S_0 = 100, \quad u = 1.1, \quad d = 0.9, \quad R = 1.02, \quad K = 100.$$

The risk-neutral up probability is

$$p = \frac{R - d}{u - d} = 0.6.$$

At maturity, the stock values and put payoffs are

S_3	133.1	108.9	89.1	72.9
$(K - S_3)^+$	0	0	10.9	27.1

NO-ARBITRAGE At $S_2 = 81$, $19 > \big[0.6(10.9) + 0.4(27.1)\big]/1.02 = 17.04$, so exercise.

The early-exercise premium comes from node-by-node choices

Working backward gives:

Node	Exercise	Continuation	Action
$t = 2,\ S = 81$	19.00	17.04	*exercise*
$t = 1,\ S = 90$	10.00	9.97	*exercise*
$t = 0,\ S = 100$	0.00	4.91	*continue*

$$V_0^{\mathrm{Eur}} \approx 4.59, \qquad V_0^{\mathrm{Am}} \approx 4.91.$$

YOUR TURN What is the early-exercise premium?

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YOUR TURN What is the early-exercise premium?

Answer

$4.91 - 4.59 = 0.32$: the value of exercising at low-stock nodes.

Numerical habits complete the toolkit

Euler simulation is simple, but its approximation must be named

For

$$\mathrm{d}X_t = b(X_t, t)\,\mathrm{d}t + a(X_t, t)\,\mathrm{d}W_t,$$

the Euler scheme on a grid is

$$\boxed{X_{n+1} = X_n + b(X_n, t_n)\Delta t + a(X_n, t_n)\sqrt{\Delta t}\, Z_{n+1},}$$

where $Z_{n+1} \sim N(0, 1)$.

What is approximated

The continuous diffusion and any path functional evaluated between grid dates.

What must be checked

Convergence in Δt, sampling error in M, positivity or boundary behavior, and bias for extrema or stopping rules.

Before trusting a numerical price, identify its weakest link

Problem	First diagnostic
Asian Monte Carlo	refine the integration grid and estimate sampling error
Barrier Monte Carlo	test monitoring bias and barrier correction
American tree	refine time steps and map the exercise region
PDE or local-vol grid	test boundary placement and mesh convergence

NO-ARBITRAGE A numerical result is credible only when it survives a refinement or a no-arbitrage check appropriate to the contract.

Takeaways

The same pricing logic adapts to the contract

Contract feature	Mathematical tool	Extra state or decision
Nonlinear Brownian function	Itô formula	quadratic variation
Drifted barrier event	Girsanov and reflection	hitting probability
Asian average	state augmentation	running integral I_t
American exercise	dynamic programming	stop or continue

NO-ARBITRAGE The payoff determines which information must be carried forward: a state variable, a stopping rule, or a path-monitoring convention.

Final check: choose the representation before calculating

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Answer

State the risk-neutral dynamics, whether monitoring is continuous or discrete, and the numerical approximation used for the hitting event. These choices determine the probability being priced.

Good pricing begins by identifying the right state, measure, and event.